HACKHARVARD 2026 · REGENERON: REINVENT CLINICAL TRIALS

TrialMate

A texting companion for clinical trials. It finds trials, keeps participants on track, and brings in a human for anything medical.

Egor Nikotin

THE PROBLEM

Trials stall on people, not science

11%

of sites in a typical trial enroll no patients at all

37%

of sites enroll fewer patients than planned

~2×

sponsors nearly double their planned timelines to hit enrollment

And between visits, participants are on their own: questions wait for office hours, side effects get forgotten, and coordinators chase people by phone.

Source: Tufts Center for the Study of Drug Development (2013), as reported by FierceBiotech

THE GAP

Participants live in their texts. Studies don’t.

Moment	Today	With TrialMate
Finding a trial	Dense registry pages and eligibility jargon	Text your condition and city, get plain-language recruiting matches and a quick pre-screen
A question at 9 pm	Wait for office hours, or guess	Answered from the study’s approved protocol, any time
A new side effect	Often forgotten by the next visit	Logged and graded on the spot; urgent ones go straight to the coordinator
Visit reminders	Coordinator phone calls	Scheduled reminders and check-ins by text

THE SOLUTION

One texting agent, three jobs

Find

Live recruiting trials from ClinicalTrials.gov, explained simply, with a yes/no pre-screen that becomes a referral.

Support

Visit questions answered from the protocol, reminders, dose check-ins, and side effects logged with a grade.

Escalate

Anything medical goes to a coordinator’s approval queue. Emergencies get 911 or 988 instantly.

Runs on Claude with role-gated tools. Works over iMessage through Photon, or the dashboard’s phone simulator.

HOW IT WORKS

One real conversation

When is my next visit?

Your next visit is the Week 4 visit on Tuesday, Oct 20 at 10:00 AM, at Cambridge Research Clinic, 3rd floor.

Since my shot yesterday I've had a headache and the spot on my arm is red and itchy

Draft, waiting for coordinator: "I've logged both the headache and the red, itchy spot for the study team…"

get_my_study_info

Reads this participant's own visits. Sent right away: routine question.

log_symptom × 2

Headache and injection-site reaction, each graded 1–4 for the team.

Held for approval

Medical turn: the participant gets a short acknowledgement, a coordinator approves the reply.

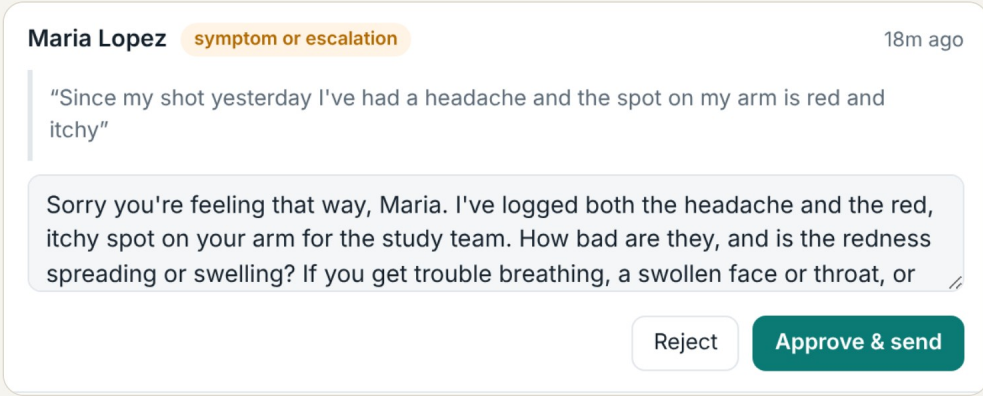
Texts from a live run with Claude on the TrialMate dashboard (fictional demo study)

SAFETY BY DESIGN

The AI never practices medicine

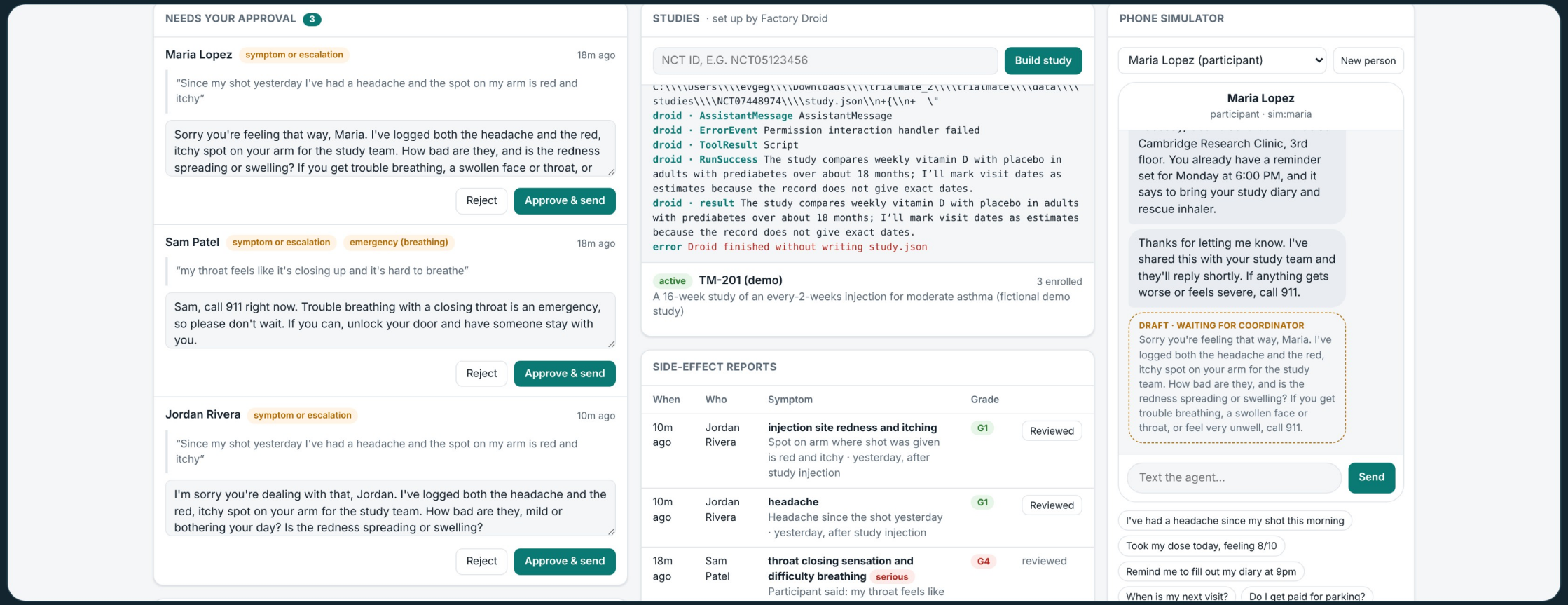
Remove secrets ▷ Emergency check ▷ Claude + tools ▷ Approval gate ▷ Outbox

- **Emergencies:** plain code sends 911 or 988 before any AI runs, and alerts the team
- **Never:** dosing advice, diagnoses or unblinding; drafts that look like them are held
- **Privacy:** SSNs, card numbers and passwords are removed before the model sees them
- **Identity** comes from the phone channel, so tools only touch the sender's own record



FOR THE STUDY TEAM

One console for the whole study



Approvals · escalations · graded side effects · dose adherence · referrals · studies · phone simulator

FACTORY DROID SDK

Droid sets up each new trial

1. A coordinator types a ClinicalTrials.gov ID on the dashboard
2. TrialMate saves the official record to one study folder and starts a Droid session (Python SDK, low autonomy)
3. Droid writes study.json: FAQ, visit schedule, screening questions, red-flag symptoms
4. Our check_study tool validates it; Droid fixes what fails until it passes
5. TrialMate re-checks it and loads a draft; a coordinator activates it

Droid writes	Used in the live product
Screening questions	Pre-screen for people who pick the trial
Visit schedule	Each enrolled participant's visits and reminders
Red-flag symptoms	Every participant text is checked; a match escalates
FAQ	Becomes the protocol the agent answers from

Code: trialmate/study_builder.py · permission requests answered by TrialMate: edits inside the study folder only

READINESS

Runs today. Ready for a one-site pilot.

What works now

- Python standard library + SQLite: starts with one click on a laptop
- Live ClinicalTrials.gov data, Claude via the Anthropic API
- Photon iMessage API connection and a dashboard phone simulator
- Write-before-send outbox with idempotency keys and retries
- MIT licensed, 30 tests, 35-message eval suite

Pilot plan

- One site, one study, coordinator approves every medical reply
- Measure: time to answer, time to side-effect report, missed visits, dropout
- Next: IRB-approved message templates, EHR and EDC export, Spanish

TRIALMATE

Every message gets an answer. Every medical one gets a human.

github.com/egornikot/trialmate

Team TrialMate · [team member names] · HackHarvard 2026